

exp3_id3.py - E:/machine_learning_lab_exp/exp3_id3.py (3.10.11)

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```
from sklearn.tree import DecisionTreeClassifier
```

```
# Input data (already encoded)
```

```
# Outlook, Temperature, Humidity, Wind
```

```
X = [  
    [0, 0, 0, 0], # Sunny, Hot, High, Weak  
    [0, 0, 0, 1], # Sunny, Hot, High, Strong  
    [1, 0, 0, 0], # Overcast, Hot, High, Weak  
    [2, 1, 0, 0], # Rain, Mild, High, Weak  
    [2, 2, 1, 0], # Rain, Cool, Normal, Weak  
    [2, 2, 1, 1] # Rain, Cool, Normal, Strong  
]
```

```
# Output (PlayTennis)
```

```
# No = 0, Yes = 1
```

```
y = [0, 0, 1, 1, 1, 0]
```

```
# Create ID3 model
```

```
model = DecisionTreeClassifier(criterion='entropy')
```

```
model.fit(X, y)
```

```
# New sample to classify
```

```
new_sample = [[1, 2, 1, 0]] # Overcast, Cool, Normal, Weak
```

```
result = model.predict(new_sample)
```

```
print("Prediction:", result)
```

IDLE Shell 3.10.11

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```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023  
, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32  
Type "help", "copyright", "credits" or "license()"   
for more information.
```

```
>>>
```

```
===== RESTART: E:/machine_learning_lab_e  
xp/exp3_id3.py =====  
Prediction: [1]
```

```
>>>
```

Ln: 6 Col: 0

Ln: 7 Col: 45



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